

hyper2kvm Migration Demo

VMware Photon OS 5.0 VMDK to KVM — End-to-End

Live demonstration: source VMDK inspection, YAML configuration, automated migration pipeline (flatten, offline fixes, initramfs rebuild, fstab repair, GRUB update), libvirt domain XML generation, VM boot on KVM, and network connectivity verification.

Photon OS 5.0 | VMDK 996 MB | QCOW2 | VirtIO | Q35/UEFI | Libvirt | SSH Verified

Step 1-2: Source & Configuration

STEP 1 — Source VMDK

```
$ ls -lh photon.vmdk
-rw-r--r-- 1 qemu qemu 996M Mar 21 11:45 photon.vmdk

$ qemu-img info photon.vmdk
image: photon.vmdk
file format: vmdk
virtual size: 8 GiB (8589934592 bytes)
disk size: 995 MiB
cluster_size: 65536
Format specific information:
  create type: monolithicSparse
  extents:
    [0]:
      virtual size: 8589934592
      filename: photon.vmdk
      cluster size: 65536
```

STEP 2 — Migration YAML Config

```
cmd: local
vmdk: ./photon.vmdk
output_dir: ./out-photon-demo
flatten: true
flatten_format: qcow2
to_output: photon.qcow2
out_format: qcow2
compress: false
fstab_mode: stabilize-all
regen_initramfs: true
emit_domain_xml: true
vm_name: photon-demo
memory: 2048
vcpus: 2
machine: q35
libvirt_test: true          # Boot VM after conversion
timeout: 60                 # Wait 60s for boot
```

```
keep_domain: true          # Keep VM running after test
```

Single command: `sudo h2kvmctl --config photon-demo.yaml` — handles everything: flatten, offline fixes, convert, generate XML, boot, verify.

Step 3: Migration Pipeline

STEP 3a — Sanity Checks (passed)

```
17:55:43 Sanity check 1/5: args...
17:55:43 Sanity check 2/5: tools...
17:55:43 Sanity check 3/5: disk space...
17:55:43 Sanity check 4/5: permissions...
17:55:43 Sanity check 5/5: network...
17:55:43 Sanity: OK
17:55:43 ✓ Sanity checks passed
```

STEP 3b — Flatten VMDK to QCOW2

```
17:55:45 Flatten snapshots -> out-photon-demo/work
17:55:45 Flattening via qemu-img convert (single self-contained image)
17:55:45 Flattening progress: 0.8%
17:55:46 Flattening progress: 5.9%
17:55:46 Flattening progress: 11.3%
17:55:46 ✓ Flattened: working-flattened-20260324.qcow2
```

STEP 3c — Offline Guest Fixes (VMCraft Engine)

```
17:55:47 Opening offline image via qemu-nbd
17:55:47 Image validated: qcow2 (virtual: 8.00 GiB, actual: 0.97 GiB)
17:55:47 Connected to /dev/nbd0
17:55:47 Detected Linux OS: VMware Photon OS 5.0
17:55:47   Product: VMware Photon OS/Linux
17:55:47   Distribution: photon
17:55:47   Version: 5.0
17:55:47   Kernel: 6.1.10-11.ph5
17:55:48 fstab scan: total_lines=4 entries=3 bypath_entries=0
17:55:48 Starting network fixes
17:55:48 Masked: vmtoolsd.service
17:55:48 Masked: vgauthd.service
17:55:48 Detected boot mode: UEFI
17:55:48 Setting kernel cmdline root=UUID=311182bd-...
17:55:51 Running initramfs rebuild: dracut -f --kver 6.1.10-11.ph5
```

```
--add-drivers virtio_blk virtio_scsi virtio_net nvme
```

STEP 3d — Convert to Final QCOW2

```
17:55:52 Convert image -> photon.qcow2
17:55:52 Converting: working-flattened.qcow2 -> photon.qcow2
      (in_format=qcow2, out_format=qcow2, compress=False)
17:55:52 Conversion progress: 3.9%
17:55:53 Conversion progress: 12.0%
17:55:53 ✓ Validated: photon.qcow2
```

Step 4: Libvirt Boot & Verification

STEP 3e — Domain XML Generation & Boot Test

```
17:55:53 emit_domain_xml: guest=linux uefi=True name=photon-demo
17:55:53 Domain XML: out-photon-demo/libvirt/photon-demo.xml
17:55:53 NVRAM: photon-demo.VARS.fd
```

```
Libvirt smoke test
Domain: photon-demo
Disk: out-photon-demo/photon.qcow2
Guest: linux
Machine: q35 | Firmware: UEFI | Network: default
Domain reached RUNNING state after 0s.
✓ Smoke test passed: domain is RUNNING
```

STEP 4 — VM Running on KVM

virsh dominfo photon-demo

```
Name:          photon-demo
State:         running
CPU(s):        2
Max memory:    2097152 KiB
Machine:       pc-q35-10.1
Firmware:      UEFI (OVMF)
Disk bus:      VirtIO (vda)
Network:       VirtIO (virbr0)
```

VM Console Screenshot

```
Welcome to Photon 5.0 (x86_64) - Kernel 6.1.10-11.ph5 (tty1)
photon-2e2948360ed5 login:
```

STEP 5 — Network Connectivity

```
$ virsh domiflist photon-demo
Interface  Type      Source      Model      MAC
-----
vnet3      network   default     virtio      52:54:00:6b:54:c1

$ virsh domifaddr photon-demo
Name      MAC address      Protocol      Address
-----
vnet3      52:54:00:6b:54:c1  ipv4          192.168.122.242/24

$ nc -zv 192.168.122.242 22
Ncat: Connected to 192.168.122.242:22.
✓ SSH port 22 OPEN – VM is network-accessible

$ arp -n | grep 192.168.122.242
192.168.122.242  ether  52:54:00:6b:54:c1  C  virbr0
```

Step 6: Output & Summary

STEP 6 — Generated Output Files

out-photon-demo/		
photon.qcow2	1016 MB	Final converted image (VirtIO, UEFI)
photon-demo.VARS.fd	128 KB	UEFI NVRAM variables
libvirt/		
photon-demo.xml	1.8 KB	Libvirt domain XML (Q35, VirtIO, OVMF)
work/		
working-flattened.qcow2	1013 MB	Intermediate flattened image

Generated Libvirt Domain XML

```
<domain type='kvm'>
  <name>photon-demo</name>
  <memory unit='MiB'>2048</memory>
  <vcpu>2</vcpu>
  <os>
    <type arch='x86_64' machine='q35'>hvm</type>
    <loader type='pflash'>/usr/share/edk2/ovmf/OVMF_CODE.fd</loader>
    <nvram template='OVMF_VARS.fd' />
  </os>
  <cpu mode='host-passthrough' />
  <devices>
    <disk type='file'>
      <driver name='qemu' type='qcow2' cache='none' io='native' discard='unmap' />
      <source file='photon.qcow2' />
      <target dev='vda' bus='virtio' />
    </disk>
    <interface type='network'>
      <source network='default' />
      <model type='virtio' />
    </interface>
    <serial type='pty' />
    <graphics type='spice' />
    <rng model='virtio'><backend>/dev/urandom</backend></rng>
    <memballoon model='virtio' />
  </devices>
</domain>
```


Migration Pipeline Summary

#	Step	Action	Duration	Result
1	Sanity checks	Tools, disk space, permissions	<1s	5/5 passed
2	Flatten	VMDK → QCOW2 (qemu-img)	~2s	working-flattened.qcow2
3	Offline fixes	fstab, GRUB, initramfs, VMware cleanup	~8s	VirtIO drivers, UEFI, UUID-based fstab
4	Convert	Working → final QCOW2	~1s	photon.qcow2 (1016 MB)
5	Emit XML	Generate libvirt domain	<1s	Q35, VirtIO, OVMF, Spice
6	Boot test	virsh define + start	<1s	RUNNING state confirmed
7	Network	DHCP lease, SSH check	~10s	192.168.122.242, SSH port 22 open

Migration Complete — Zero Manual Intervention

VMware Photon OS 5.0 VMDK (996 MB) → KVM QCOW2 with VirtIO + UEFI + Q35
Offline fixes: initramfs rebuilt, fstab stabilized, VMware tools masked, GRUB updated
VM booted on KVM, reached login prompt, SSH accessible on 192.168.122.242:22
Total time: ~12 seconds. Single command. Zero BSOD. Zero manual steps.